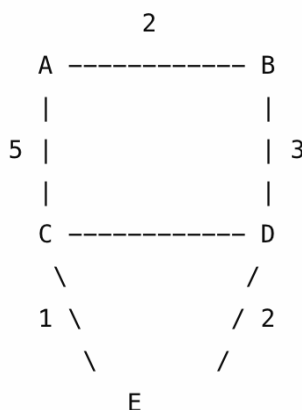


Exercise 2: Link State and Distance Vector Routing

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Learning objectives: Apply Dijkstra and Bellman-Ford; construct forwarding tables; analyze convergence, failures, and count-to-infinity.

Network topology:



Part 1 — Link State Routing

Question 1 — Dijkstra's Algorithm

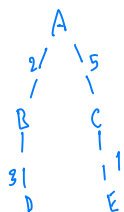
Starting at router A, execute Dijkstra's algorithm. Show every update and the node confirmed at each step.

Step	Confirmed Node	D(B), Prev.	D(C), Prev.	D(D), Prev.	D(E), Prev.
0	A	2, A	5, A	∞	∞
1	B	2, A	5, A	5, B	∞
2	C			5, B	6, C
3	D			5, B	6, C
4	E				6, C

Question 2 — Shortest-Path Tree

Draw the shortest-path tree rooted at A, then list the shortest path and total cost.

SPT sketch:



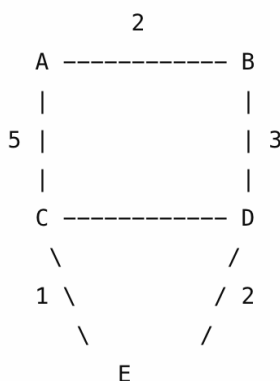
Destination	Shortest Path	Total Cost
B	A → B	2
C	A → C	5
D	A → B → D	5
E	A → C → E	6

Question 3 — Forwarding Table

Destination	Shortest Cost	Next Hop	Complete Path
B	2	B	A→B
C	5	C	A→C
D	5	B	A→B→D
E	6	C	A→C→E

Part 2 — Distance Vector Routing

Consider the same network. Initially, each router knows only the cost to its directly connected neighbors.



Question 4 — Bellman-Ford Update

Router A receives the following distance vectors from its neighbors:

From B:

A=2, B=0, C=4, D=3, E=5;

From C:

A=5, B=4, C=0, D=1, E=1.

Use the Bellman-Ford equation. Calculate router A's new distance to:

Show all calculations, including the neighbor through which the minimum-cost route is obtained.

Destination	Via B calculation	Via C calculation	Minimum Cost	Next Hop
B	$2 + D-B(B) = 2 + 0 = 2$	$5 + D-C(B) = 5 + 4 = 9$	2	B
C	$2 + D-B(C) = 2 + 4 = 6$	$5 + D-C(C) = 5 + 0 = 5$	5	C
D	$2 + D-B(D) = 2 + 3 = 5$	$5 + D-C(D) = 5 + 1 = 6$	5	B
E	$2 + D-B(E) = 2 + 5 = 7$	$5 + D-C(E) = 5 + 1 = 6$	6	C

Question 5 — Distance Vector Table

After receiving the distance vectors, construct router A's updated routing table.

Destination	Next Hop	Distance	Full Path
B	B	2	A→B
C	C	5	A→C
D	B	5	A→B→D
E	C	6	A→C→E

Part 3 — Routing Convergence

Question 6 — Link Failure

The C-D link fails. Explain separately what happens with Link State and Distance Vector routing.

Link State routing.

Your answer should describe:

- Which router(s) detect the failure.
- What information is flooded through the network.
- How routers update their link-state databases.
- How routers calculate new shortest paths.

Distance Vector routing.

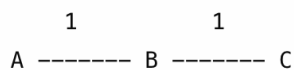
Your answer should describe:

- How routers learn about the failure.
- How updated distance vectors propagate.
- Why routers may temporarily have incorrect routing information.
- How routing eventually converges.

Link State	Distance Vector
C and D detect the failure	Router learn about failure through update distance vector receive from neighbor
C,D generate LSA and flood it on every interface except arrived on	Advertise new distance to neighbor routers hop by hop
Every router compare sequence number	Router may still believe that destination is reachable through another neighbor
All routers calculate new routes without using failed C-D link	Router recalculate distance using Bellman-Ford Equation and reach stable state and converges.

Part 4 — Count-to-Infinity

Consider



Initially A reaches C through B at cost 2. The B–C link fails, but B receives stale information from A saying C is reachable at cost 2.

Question 7

- [illegible]

Part 5 — Compare LS and DV

Question 8

Characteristic	Link State	Distance Vector
Basic algorithm	Dijkstra	Bellman-Ford
Information maintained	Link-state topology database	Distance and next-hop
Information exchanged	Link-state advertisements are flooded	DV are exchanged with neighbor
Algorithm	Each router calculates shortest paths	Calculate minimum cost
Topology knowledge	Know topology in routing area	Don't know complete topology
Routing updates	Flooded when topology changes	sent/update between neighbor
Convergence	Fast	Slower
Routing loops	Less prone after synchronized topology	More susceptible to temporary loops
Count-to-infinity	Not a typical LS problem	Major DV problem
Example protocol	OSPF	RIP

Part 6 — Design Question

Question 9

You are designing a routing system for an enterprise with 500 routers requiring fast convergence, accurate topology knowledge, efficient routing, and hierarchical design. Choose LS or DV and justify your choice with at least three technical arguments. Your answer should discuss at least three advantages and/or disadvantages of your chosen approach.

Link state info:

1. Convergence is much faster because topology information allows shortest path calculation
2. Accurate topology knowledge using Router's link-state database
3. Hierarchical design like OSPF's concept of areas
4. Avoids count-to-infinity problem

Challenge — Algorithmic Analysis

Question 10

For N routers and E links, compare LS and DV in information maintained, communication overhead, computation, convergence, and scalability. Explain why large-scale intra-domain routing generally favors Link State.

	Linkstate	Distance vector
Information maintained	maintain topology/link state db	maintain distance, hops to destination
Communication overhead	flooded through when change occur	exchange between neighbor
Computation	$O(E \log N)$ with priority queue	require multiple iterations
Convergence	fast after synchronized	slower after failures
Scalability	OSPF area	Difficult to scale very large networks

Link state is faster because router can maintain consistent topology db and independently calculate shortest path
 Link state protocol can also use hierarchy structures such as areas to limit the scope of topology information and routing calculation